

Class:

Name:

Date:



Grade 7 Science Worksheet: Exploring Waves

Part A: Vocabulary Matching

Draw a line to match each term on the left with its correct definition on the right.

Term	Definition
1. Medium	A. The distance between two matching points on a wave (e.g., crest to crest).
2. Crest	B. The number of wave cycles that pass a point in one second (measured in Hertz).
3. Trough	C. The material (solid, liquid, or gas) that a wave travels through.
4. Wavelength (λ)	D. The highest point of a wave.
5. Amplitude	E. The lowest point of a wave.
6. Frequency (f)	F. The height of a wave, which relates to its energy or loudness.

Part B: Mechanical vs. Electromagnetic Waves

Fill in the blanks using words from the word bank below. (Some words may be used more than once).

Word Bank: Solid, Vacuum, Electromagnetic, Sound, Light, Liquid, Gas, **Mechanical waves**

- Waves that require a physical medium to travel through are called _____ waves. An example of this is a _____ wave.
- Waves that do **not** need a medium and can travel through the empty space of a _____ are called _____ waves. An example of this is _____.
- Mechanical waves travel **fastest** in a _____ because the particles are packed very tightly together. They travel **slowest** in a _____.

Class:

Name:

Date:



Part C: Drawing Wave Anatomy

Using a pencil, draw a simple transverse wave in the box below. Label the following four parts on your drawing:

- **Equilibrium line** (the resting center line)
- **Crest**
- **Trough**
- **Wavelength**
- **Amplitude**



Part D: Wave Behaviors & The Real World

Answer the following short-answer questions based on how waves interact with the world around them.

1. **Active Noise Cancellation (ANC):** Imagine you are wearing noise-canceling headphones. If a noisy lawnmower makes a wave with a high "crest," what does your headphone's microphone generate to cancel it out? What is this process called?

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2. **The Doppler Effect:** You are standing on a sidewalk when an ambulance drives past you with its sirens blaring.

As the ambulance drives **towards** you, do the sound waves squish together (higher pitch) or stretch out (lower pitch)?

As the ambulance drives **away** from you, what happens to the sound of the siren?



Class:

Name:

Date:

3. **Cosmic Colors (Stars):** Space scientists look at the temperature of stars by studying their colors.

A **Blue Star** is _____ (hotter / cooler) than a **Red Star**.

4. **Reflection vs. Absorption:** If you want to build a quiet recording studio, would you line the walls with **hard concrete walls** or **soft foam panels**? Explain your choice using wave behaviors.

Part E: The Basic Wave Math Lab

In science, we use the **Universal Wave Equation** to see how fast a wave is moving:

$$\text{Velocity } (v) = \text{Frequency } (f) \times \text{Wavelength } (\lambda)$$

- **Velocity** (v) is measured in meters per second (m/s).
- **Frequency** (f) is measured in Hertz (Hz).
- **Wavelength** (λ) is measured in meters (m).

Solve the two problems below. Show your work!

1. A sound wave traveling through the air has a frequency of 170 Hz and a wavelength of 2 meters

. How fast is the sound wave traveling?

Formula: $v = f \times \lambda$

Show Work:

Answer: _____ m/s



Class:

Name:

Date:

2. A wave in the ocean travels at a speed of 6 m/s. If its wavelength is 3 meters, what is the frequency of the wave? (*Hint: Rearrange the formula to find f*)

Formula: $f = \frac{v}{\lambda}$

Show Work:

Answer: _____ Hz